Year 9 · Chapter 11 · Coordinate geometry and graphs: answers (teacher copy)

11.1 The length of an interval (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: $d^2 = 3^2 + 4^2 = 25$, so $d = \sqrt{25} = 5$
2. Ex 1b: $d = \sqrt{(8 - 2)^2 + (9 - 1)^2} = \sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$
3. Ex 1c: $d = \sqrt{(3 - (-1))^2 + (-1 - 2)^2} = \sqrt{(4^2 + (-3)^2)} = \sqrt{25} = 5$
4. Ex 1d: $d = \sqrt{(4 - 1)^2 + (8 - 2)^2} = \sqrt{(3^2 + 6^2)} = \sqrt{45} = 6.7$
5. Ex 2a: $d = \sqrt{(4 - (-2))^2 + (-3 - 5)^2} = \sqrt{(6^2 + (-8)^2)} = \sqrt{100} = 10$
6. Ex 2b: $d = \sqrt{(3 - (-3))^2 + (1 - (-2))^2} = \sqrt{(6^2 + 3^2)} = \sqrt{45} = 6.7$
7. Ex 2c: $AB = \sqrt{(4^2 + 3^2)} = 5$, so $P = 4 + 3 + 5 = 12$
8. Ex 2d: $PR = \sqrt{(2^2 + 4^2)} = \sqrt{20}$ and $QR = \sqrt{((-2)^2 + 4^2)} = \sqrt{20}$, so PR = QR: isosceles

Easy

1. 5
2. $\sqrt{13} = 3.6$
3. $\sqrt{26} = 5.1$
4. $\sqrt{32} = 5.7$
5. 5
6. 10
7. $\sqrt{29} = 5.4$
8. $\sqrt{45} = 6.7$
9. 10
10. 13
11. 5
12. $\sqrt{52} = 7.2$
13. $\sqrt{72} = 8.5$
14. $\sqrt{32} = 5.7$
15. $\sqrt{34} = 5.8$
16. $\sqrt{41} = 6.4$
17. $\sqrt{45} = 6.7$
18. 8
19. 7
20. 5

Medium

1. 10
2. 15 cm (Ch 2)
3. 10
4. $\sqrt{45} = 6.7$
5. $x = 7$ (Ch 7)
6. $6 + 8 + 10 = 24$ units
7. $OA = \sqrt{34} = 5.8$, $OB = \sqrt{32} = 5.7$, so B
8. $V = \pi \times 3^2 \times 10 = 282.7$ cm³ (10.10)
9. $(k - 1)^2 + 4^2 = 25$, so $k - 1 = 3$ and $k = 4$

Worded problems

1. $d = \sqrt{(5 - (-3))^2 + (8 - 2)^2} = \sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$ units, so 1 000 m
2. $d = \sqrt{(14 - 2)^2 + (6 - 1)^2} = \sqrt{(12^2 + 5^2)} = \sqrt{169} = 13$ km; road $12 + 5 = 17$ km, so $17 - 13 = 4$ km shorter
3. $AC = \sqrt{(3^2 + 4^2)} = 5$, $BC = \sqrt{((-3)^2 + 4^2)} = 5$, AB = 6: AC = BC, so isosceles. $P = 5 + 5 + 6 = 16$ units

11.2 The midpoint of an interval

(booklet p. 6)

WE DO examples (teacher)

1. Ex 1a: $M = ((2 + 8) \div 2, (4 + 10) \div 2) = (5, 7)$
2. Ex 1b: $M = ((-3 + 5) \div 2, (5 + 1) \div 2) = (1, 3)$
3. Ex 1c: $M = ((-4 + 2) \div 2, (-3 + 1) \div 2) = (-1, -1)$
4. Ex 1d: $x: 2 \times 4 - 1 = 7$, $y: 2 \times 5 - 3 = 7$, so B = (7, 7)
5. Ex 2a: $M = ((-4 + 6) \div 2, (7 + (-3)) \div 2) = (1, 2)$
6. Ex 2b: $x: 2 \times 1 - (-3) = 5$, $y: 2 \times (-1) - 2 = -4$, so B = (5, −4)
7. Ex 2c: The centre is the midpoint of the diameter: $M = ((-2 + 4) \div 2, (-1 + 5) \div 2) = (1, 2)$
8. Ex 2d: $(k + 8) \div 2 = 5$, so $k + 8 = 10$ and $k = 2$

Easy

1. (4, 6)
2. (4, 4)
3. (6, 4)
4. (5, 4)
5. (2.5, 5)
6. (4, 4.5)
7. (4.5, 2.5)
8. (6, 7.5)
9. (2, 1)
10. (−2, 2)
11. (−4, −2)
12. (0, 1)
13. (−1, 1)
14. (−1, 0)
15. (0.5, −1.5)
16. (2.5, −1.5)
17. (1.5, −2.5)
18. (7, 8)
19. (4, −1)
20. (−3, 2)

Medium

1. (2, 3)
2. 10 (11.1)
3. (10, 3)
4. (0, 0.5)
5. $x = 10$ (Ch 7)
6. $(k + 8) \div 2 = 3$, so $k = -2$
7. (−3.5, −4.5)
8. $A = \pi \times 5^2 = 78.5$ cm² (10.7)
9. $M = (4, 5)$; $d = \sqrt{(4 - 7)^2 + (5 - 2)^2} = \sqrt{((-3)^2 + 3^2)} = \sqrt{18} = 4.2$

Worded problems

1. $M = ((-4 + 8) \div 2, (3 + (-1)) \div 2) = (2, 1)$
2. Centre: $M = ((-1 + 7) \div 2, (2 + 8) \div 2) = (3, 5)$; diameter $d = \sqrt{(7 - (-1))^2 + (8 - 2)^2} = \sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$, so the radius is 5
3. Tap: $M = ((2 + 10) \div 2, (1 + 7) \div 2) = (6, 4)$; $d = \sqrt{(6 - 2)^2 + (4 - 1)^2} = \sqrt{(4^2 + 3^2)} = \sqrt{25} = 5$ m

11.3 The gradient of an interval (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: m = rise ÷ run = 2 ÷ 4 = ½
2. Ex 1b: m = −6 ÷ 3 = −2 (down to the right)
3. Ex 1c: m = (10 − 2) ÷ (5 − 1) = 8 ÷ 4 = 2
4. Ex 1d: m = 0 ÷ 7 = 0 (horizontal)
5. Ex 2a: m = (−2 − 4) ÷ (5 − (−3)) = −6 ÷ 8 = −¾
6. Ex 2b: m = 6 ÷ 4 = 3/2
7. Ex 2c: (k − 2) ÷ 3 = 3, so k − 2 = 9 and k = 11
8. Ex 2d: m = −3: it changes by 3 for each 1 across (the sign only gives the direction)

Easy

1. 2
2. ½
3. 1
4. 3
5. −2
6. ½
7. −1
8. 3
9. 2
10. −2
11. ½
12. 3
13. −⅓
14. −⅔
15. 0 (zero: horizontal)
16. undefined (vertical)
17. −⅔
18. 2
19. −1
20. ½

Medium

1. 2
2. (2, 3) (11.2)
3. −3/2
4. −2
5. 10 (11.1)
6. Undefined: the run is 3 − 3 = 0 (a vertical line)
7. m = 1.2 ÷ 4.8 = 0.25 (¼)
8. V = 8 × 5 × 3 = 120 cm³ (10.10)
9. (11 − k) ÷ 3 = 3, so 11 − k = 9 and k = 2

Worded problems

1. m = 45 ÷ 600 = 0.075, which is 7.5%
2. Roof: m = 2.4 ÷ 4 = 0.6. Ramp: 0.5. 0.6 > 0.5, so yes, the roof is steeper
3. m(AB) = 4 ÷ 2 = 2 and m(BC) = 6 ÷ 3 = 2: the same gradient through the shared point B, so A, B and C are on one line

11.4 Graphing linear equations (booklet p. 12)

WE DO examples (teacher)

1. Ex 1a: y: −1, 1, 3, 5
2. Ex 1b: y = 0: 0 = −3x + 6, so x = 2 → (2, 0); x = 0: y = 6 → (0, 6)
3. Ex 1c: y = 4 − 6 = −2 ✓ Yes, (4, −2) is on the line
4. Ex 1d: x = 3: a vertical line through 3 on the x-axis; y = −2: a horizontal line through −2 on the y-axis
5. Ex 2a: Intercepts (2, 0) and (0, 4); rule a line through them
6. Ex 2b: 4 − 3 × (−2) = 4 + 6 = 10 ✓ Yes
7. Ex 2c: y = 0: 3x = 12, x = 4 → (4, 0); x = 0: 4y = 12, y = 3 → (0, 3)
8. Ex 2d: x = −2, x = 3, y = −1, y = 3: a 5 by 4 rectangle, A = 20 square units

Easy

1. y: 3, 5, 7, 9
2. y: −2, −1, 0, 1
3. y: 0, 3, 6, 9
4. y: 4, 3, 2, 1
5. (2, 0) and (0, −4)
6. (3, 0) and (0, 3)
7. (−2, 0) and (0, 6)
8. (4, 0) and (0, 8)
9. y: −3, −1, 1, 3
10. y: 5, 3, 1, −1
11. y: 1, 2, 3, 4
12. Yes: 2 × 3 − 1 = 5
13. No: −3 × (−2) − 1 = 5, not 4
14. Yes: 4 − 7 = −3
15. (2, 0) and (0, −8)
16. (3, 0) and (0, 6)
17. (4, 0) and (0, −2)
18. x = −2
19. y = 3
20. y = −1

Medium

1. (−5, 0) and (0, 10)
2. −2 (11.3)
3. Yes: −2 × (−3) + 1 = 7
4. (−2, 0) and (0, 4)
5. 10 (11.1)
6. y = −3
7. (4, 0) and (0, −10)
8. (−2, −2) (11.2)
9. Intercepts (3, 0) and (0, 6): A = ½ × 3 × 6 = 9 square units

Worded problems

1. C: 40, 100, 160, 220. C-intercept 40: the $40 call-out fee (charged for 0 hours)
2. (6, 0) and (0, 9); d = √(6² + 9²) = √117 = 10.8
3. Width 5 − (−1) = 6, height 2 − (−3) = 5: P = 22 units, A = 30 square units

11.5 The gradient–intercept equation

(booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: m = 3, c = 1
2. Ex 1b: y = −x + 8: m = −1, c = 8
3. Ex 1c: m = ⅕, c = 4
4. Ex 1d: Plot (0, −3); from it go right 1, up 2 to (1, −1); rule the line
5. Ex 2a: 2y = −3x + 12, so y = −1.5x + 6: m = −1.5, c = 6
6. Ex 2b: Plot (0, 4); right 1, down 2 to (1, 2); rule the line
7. Ex 2c: y = −3x + 5
8. Ex 2d: Parallel, so m = 4: y = 4x + 3

Easy

1. m = 4, c = 3
2. m = −2, c = 5
3. m = 1, c = −7
4. m = ½, c = 1
5. m = −3, c = 6
6. m = −1, c = 0
7. m = 0, c = 9
8. m = 0.5, c = −2
9. y = −2x + 7: m = −2, c = 7
10. y = 3x + 4: m = 3, c = 4
11. y = −2x + 5: m = −2, c = 5
12. (0, −1) and (1, 1)
13. (0, 3) and (1, 2)
14. (0, 1) and (2, 2)
15. y = 3x − 2
16. y = −½x + 4
17. y = 5
18. Yes: both have m = 2
19. No: m = 3 and m = −3
20. Yes: both have m = −1

Medium

1. m = −6, c = 0
2. x = 4 (11.4)
3. y = −2x + 3
4. y = −x + 2
5. 2 (11.3)
6. y = ¾x − 2
7. 4x − 2y = 6 gives y = 2x − 3: both have m = 2, so yes
8. (1, 3) (11.2)
9. y = −3x − 4

Worded problems

1. Gradient 0.5: $0.50 for each gigabyte. C-intercept 15: the $15 monthly fee (0 GB used)
2. m = 2, c = 3 − 5 = −2: y = 2x − 2. x-intercept: 0 = 2x − 2, x = 1
3. They did not divide every term by 2. 2y = 6x + 8, so y = 3x + 4: m = 3, c = 4

11.6 Finding the equation of a line

(booklet p. 18)

WE DO examples (teacher)

1. Ex 1a: c = 1, m = 2 ÷ 1 = 2: y = 2x + 1
2. Ex 1b: c = 2, m = −3 ÷ 1 = −3: y = −3x + 2
3. Ex 1c: c = −1, m = 1 ÷ 2 = ½: y = ½x − 1
4. Ex 1d: c = −2, m = −1: y = −x − 2
5. Ex 2a: c = 2, m = −2 ÷ 3: y = −⅔x + 2
6. Ex 2b: m = −9 ÷ 3 = −3; 7 = −3 × (−1) + c, so c = 4: y = −3x + 4
7. Ex 2c: −6 = 4 × (−1) + c, so c = −2: y = 4x − 2
8. Ex 2d: 4 × 3 − 2 = 10 ✓ Yes

Easy

1. y = x + 2
2. y = 2x − 1
3. y = −x + 1
4. y = 3x
5. y = −2x + 3
6. y = ½x + 1
7. y = −3x − 1
8. y = x − 3
9. y = 3x − 2
10. y = −½x + 2
11. y = −2x − 1
12. m = 2, c = 3: y = 2x + 3
13. m = ½, c = −1: y = ½x − 1
14. m = −3, c = 4: y = −3x + 4
15. m = 2, c = 3: y = 2x + 3
16. m = 3, c = −5: y = 3x − 5
17. m = −2, c = 2: y = −2x + 2
18. c = 5 − 3 × 2 = −1: y = 3x − 1
19. c = 4 + 2 × 1 = 6: y = −2x + 6
20. c = 1 − ½ × 4 = −1: y = ½x − 1

Medium

1. y = 4x − 5
2. m = −2, c = 7 (11.5)
3. y = 3x − 2
4. y = −x + 3
5. 4 (11.4)
6. y = ½x + 4
7. y = −3x + 5
8. 10 (11.1)
9. m = 2; 4 = 2 × (−1) + c, c = 6: y = 2x + 6

Worded problems

1. m = (18 − 24) ÷ 2 = −3, c = 24: h = −3t + 24. 0 = −3t + 24, so t = 8 hours
2. m = (25 − 13) ÷ (10 − 4) = 2; 13 = 2 × 4 + c, c = 5: C = 2d + 5. Flagfall $5
3. m = 12 ÷ 6 = 2; −1 = 2 × (−2) + c, c = 3: y = 2x + 3. 0 = 2x + 3, x = −1.5: (−1.5, 0)

11.7 Solving linear equations graphically

(booklet p. 21)

WE DO examples (teacher)

1. Ex 1a: The line y = 5 meets y = 2x + 1 at x = 2
2. Ex 1b: y = −3 meets the line at x = −2
3. Ex 1c: The lines cross at (2, 3)
4. Ex 1d: 2 × 2 + 1 = 5 ✓
5. Ex 2a: y = 5 meets the line at x = 3
6. Ex 2b: (2, 3)
7. Ex 2c: The lines cross at (1, 4), so x = 1
8. Ex 2d: y = 2 × 2 − 1 = 3 ✓ and y = −2 + 5 = 3 ✓

Easy

1. x = 2
2. x = 3
3. x = 4
4. x = 2
5. x = 2
6. x = 3
7. x = 3
8. x = 2
9. x = 2
10. x = −1
11. x = 3
12. (2, 3)
13. (1, 1)
14. (2, −1)
15. x = −2
16. x = 4
17. x = 2
18. x = 1.5
19. x = −2.5
20. x = 3.5

Medium

1. x = −2
2. y = 2x + 2 (11.6)
3. x = 3
4. (1, 3)
5. (6, 0) and (0, 6) (11.4)
6. 2x + 1 = 7, x = 3: (3, 7)
7. (3, 7)
8. y = −x + 6 (11.5)
9. x = 5

Worded problems

1. They cross at n = 5: 20 + 25 = 45 and 10 + 35 = 45, so 5 lessons, $45
2. V = 15 when t = 5: 40 − 5 × 5 = 15 ✓, so after 5 minutes
3. 30 + 12w = 90 − 8w at w = 3: both have $66

11.8 Linear modelling (booklet p. 24)

WE DO examples (teacher)

1. Ex 1a: C: 3.50, 5.75, 8.00, 10.25
2. Ex 1b: Gradient 2.25: $2.25 per km. C-intercept 3.50: the flagfall
3. Ex 1c: C = 2.25x + 3.5
4. Ex 1d: C = 2.25 × 12 + 3.5 = $30.50
5. Ex 2a: About $11.38 (C = 2.25 × 3.5 + 3.5 = $11.38)
6. Ex 2b: C = 2.25 × 76 + 3.5 = $174.50
7. Ex 2c: 2.25x + 3.5 = 12.05, so 2.25x = 8.55 and x = 3.8 km
8. Ex 2d: 2.25x + 3.5 = 82.25, so 2.25x = 78.75 and x = 35 km

Easy

1. $21
2. $27
3. 150 km
4. 26 L
5. y = 3x + 5
6. y = 4x + 2
7. y = −3x + 20
8. y = ½x + 1
9. C = 2d + 4
10. C = 0.1t + 25
11. V = 15t + 200
12. C = $30 (rule C = 5n + 10)
13. V = 25 L (rule V = 40 − 5t)
14. d = 37.5 km (rule d = 15t)
15. $30 per hour; $50 call-out fee
16. grows 6 mm a week; 150 mm at the start
17. spends $25 a week; $400 at the start
18. d = 13 km
19. t = 70 texts
20. t = 20 minutes

Medium

1. C = 60h + 80
2. y = 3x + 4 (11.6)
3. C = 60 × 3.5 + 80 = $290
4. y = −2x + 12
5. (2, 3) (11.7)
6. F = 60 − 7.5d
7. 60 − 7.5d = 0, d = 8: 800 km
8. m = 6 ÷ 4 = 1.5 (11.3)
9. 50 + 10w = 20 + 15w, so w = 6 weeks

Worded problems

1. h = 30 − 2.5t; h = 30 − 12.5 = 17.5 cm; 30 − 2.5t = 0 at t = 12 hours
2. C = 20 + 0.15m; C = 20 + 18 = $38; 20 + 0.15m = 50, so m = 200 minutes
3. V = 1200 + 80t; 1200 + 80t = 5000, so 80t = 3800 and t = 47.5 minutes

11.9 Graphing parabolas (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: y: 5, 2, 1, 2, 5
2. Ex 1b: vertex (0, 0), concave down
3. Ex 1c: $y = x^2$ ($a = 1$ is closer to 0 than $a = 3$)
4. Ex 1d: Vertex (0, −3), concave up, same shape as $y = x^2$: $y = x^2 - 3$
5. Ex 2a: Table y: 0, 3, 4, 3, 0; vertex (0, 4), concave down
6. Ex 2b: $y = 2x^2$ ($a > 0$ and $c = 0$)
7. Ex 2c: $3^2 + 1 = 10$ ✓ Yes
8. Ex 2d: $x^2 - 9 = 0$, $x^2 = 9$, so $x = 3$ or $x = -3$

Easy

1. y: 4, 1, 0, 1, 4
2. y: 6, 3, 2, 3, 6
3. y: 8, 2, 0, 2, 8
4. y: −4, −1, 0, −1, −4
5. y: 1, −2, −3, −2, 1
6. y: 0, 3, 4, 3, 0
7. y: 12, 3, 0, 3, 12
8. y: −7, −1, 1, −1, −7
9. vertex (0, −2), concave up
10. vertex (0, 3), concave down
11. vertex (0, 1), concave up
12. vertex (0, 0), concave down
13. vertex (0, −6), concave up
14. vertex (0, 7), concave down
15. $y = 2x^2$ (2 is closer to 0)
16. $y = -\frac{1}{3}x^2$ ($\frac{1}{3}$ is closer to 0)
17. $y = x^2 + 4$ ($a = 1$ is closer to 0)
18. $y = x^2 + 1$
19. $y = -x^2 + 2$
20. $y = x^2 - 4$

Medium

1. vertex (0, 5), concave down
2. $m = -2$, $c = 4$ (11.5)
3. Yes: $2 \times (-2)^2 - 1 = 8 - 1 = 7$
4. $y = x^2 - 1$
5. $x = 3$ (11.7)
6. $x^2 = 16$, so $x = 4$ or $x = -4$
7. $y = 2x^2$: $a = 2$ is further from 0
8. −8 (11.4)
9. $4 + c = 1$, so $c = -3$; vertex (0, −3)

Worded problems

1. Top: vertex (0, 8), so 8 m. Ground: $-0.5x^2 + 8 = 0$, $x^2 = 16$, $x = \pm 4$, so 8 m wide
2. $y = 0.25 \times 6^2 = 9$ cm
3. $y = x^2$: 4, 1, 0, 1, 4; $y = x^2 + 3$: 7, 4, 3, 4, 7. The same shape, moved up 3: vertex (0, 3)

11.10 Parabolas in real life (booklet p. 30)

WE DO examples (teacher)

1. Ex 1a: h: 20, 15, 0
2. Ex 1b: 20: the stone starts 20 m up (the cliff's height)
3. Ex 1c: $h = -5 \times 0.5^2 + 20 = 18.75$ m
4. Ex 1d: $-5t^2 + 20 = 0$, $t^2 = 4$, $t = 2$ seconds ($t \geq 0$)
5. Ex 2a: From the table h: 80, 75, 60, 35, 0 ($t = 0$ to 4)
6. Ex 2b: $h = -5 \times 2.5^2 + 80 = 48.75$ m
7. Ex 2c: $-5t^2 + 80 = 35$, $t^2 = 9$, $t = 3$ seconds
8. Ex 2d: At $t = 2$, $h = 60$: it has fallen $80 - 60 = 20$ m

Easy

1. 25 m
2. 35 m
3. 49 cm²
4. 80 m
5. h: 45, 40, 25, 0
6. d: 0, 5, 20, 45
7. A: 1, 4, 9, 16
8. h: 9, 8, 5, 0
9. 45 m
10. 40 m
11. 25 m
12. $-5t^2 + 80 = 0$, $t^2 = 16$, $t = 4$ s
13. $-5t^2 + 45 = 25$, $t^2 = 4$, $t = 2$ s
14. $5t^2 = 125$, $t^2 = 25$, $t = 5$ s
15. $x^2 = 64$, $x = 8$ cm
16. $A = 2 \times 5^2 = 50$ cm²
17. $2x^2 = 72$, $x^2 = 36$, $x = 6$ cm
18. $-5t^2 + 20 = 0$, $t^2 = 4$, $t = 2$ s
19. $-5t^2 + 180 = 0$, $t^2 = 36$, $t = 6$ s
20. $-5t^2 + 31.25 = 0$, $t^2 = 6.25$, $t = 2.5$ s

Medium

1. 80 m
2. (0, 6) (11.9)
3. $-5t^2 + 125 = 0$, $t^2 = 25$, $t = 5$ s
4. 15 m
5. $210 (11.8)
6. $x^2 = 144$, $x = 12$ cm
7. $h = -5t^2 + 50$ (45 m, not 44 m)
8. $y = 2x + 2$ (11.6)
9. $45 - 5 = 40$ m

Worded problems

1. $-5t^2 + 20 = 0$, $t^2 = 4$, $t = 2$ s. $h = -5 \times 1.5^2 + 20 = 8.75$ m
2. Height 4 m; $-0.25x^2 + 4 = 0$, $x = \pm 4$, so 8 m wide. At $x = 1.8$: $y = 3.19$ m, which is more than 3 m, so yes
3. $x^2 = 50$, $x = \sqrt{50} = 7.1$ cm